

JUSTIN CHOO JING RUI

Email: justinchoojr@gmail.com Github: <https://github.com/Justincjr>

EDUCATION

National University of Singapore

Aug 2023 - Dec 2026 (Expected)

Bachelor of Computing in Computer Science

- Focus area in Artificial Intelligence
- Coursework: Big Data Systems for Data Science, Machine Learning, Natural Language Processing, Computer Vision, Cryptography

WORK EXPERIENCE

AI Platform Engineer Intern, Intellectual Property Office of Singapore July 2026 - Present

- Architected and deployed LLM-powered agents using AWS Bedrock AgentCore to automate trademark examinations; developed a multi-stage LLM pipeline for structured field extraction from images and a RAG-based system utilizing OpenSearch and MCP tools to identify subjects in descriptions.
- Engineered a comprehensive evaluation framework consisting of 14 Lambda-based metrics, including WER, CER, WIL and semantic similarity, while implementing automated batch processing
- Provisioned and maintained cloud infrastructure via Terraform, managing Lambda evaluators, DynamoDB state persistence, IAM policies, and AgentCore production runtime environments.
- Preprocessed, structured, and validated large-scale intellectual property datasets

Software Engineer Intern, Central Provident Fund Board, Singapore

Aug 2025 - Dec 2025

- Designed and implemented a self-service data exploration platform enabling staff to search, browse, and understand organisational datasets, improving dataset lookup time from minutes to seconds for 50+ users.
- Built a full-stack incident tracking platform used by 5–10 teams, supporting 100+ incident reports with features for case creation, updates, attachments, and activity logs. Improved cross-department traceability and decreased manual tracking effort by ~50%.
- Contributed to both frontend and backend components, integrating CI/CD pipelines, authentication, and cloud-based storage.

Software Engineer Intern, Kabam Robotics, Singapore

May 2025 - Aug 2025

- Developed adapters in C++ (ROSCPP) to integrate robot with a custom frontend javascript platform.
- Enhanced real-time video streaming by enabling robot to capture RTSP camera feeds and transmit via WebRTC (C++ & JavaScript), reducing CPU utilisation from 30% to 18% and enhancing remote monitoring performance.

Software Developer Intern, Azalea Asset Management, Singapore

May 2024 - Aug 2024

- Designed automation pipelines powered by Azure OpenAI, leveraging libraries like TypeChat and Semantic Kernel. Built a GICS code classification service using GPT models, raising accuracy from 25% to 70% and reducing analyst workload.
- Deployed an internal LLM workspace platform allowing teams to chat with documents, perform semantic search, and develop custom AI agents on top of organisational knowledge.

PROJECTS

- Architected a modular Model Context Protocol (MCP) host engine powered by LangGraph, orchestrating tool-calling across a built-in ChromaDB RAG search server, live web search.
- Built an Ultimate Tic-Tac-Toe agent in PyTorch using a hierarchical neural network, utilizing shared-weight encoders to efficiently capture local 3x3 board patterns while modeling global meta-game strategy.

- Engineered a storewide sales prediction system in PySpark using a Random Forest Regression pipeline, implemented temporal and demographic feature extraction on large-scale transaction datasets, and validated model accuracy with a MAE under 1270
- Developed a multilingual Seq2Seq model in PyTorch for multilingual letter-to-phoneme conversion, implementing a GRU-based Encoder-Decoder architecture with Multi-head Attention. Reduced the Phoneme Error Rate (PER) from 0.18 to 0.06.
- Developed a text classification pipeline to categorize German book blurbs for the GermEval 2019 task, fine-tuning a BERT-based model and implementing Parameter-Efficient Fine-Tuning via QLoRA on a LLM.
- Architected a comprehensive feature engineering pipeline for Ames Housing Dataset utilizing target encoding and log-transformations to address data skewness, while executing rigorous outlier mitigation to optimize dimensionality and maintain statistical variance.
- Built and deployed an educational MERN-stack web application on AWS EC2, implementing user authentication, and a cloud-based production environment.

SKILLS

- **Programming Languages:** Python, Java, JavaScript (Node.js/React), C++, C
- **AI/ML/Robotics:** PyTorch, ROSCPP, MCP, LangChain, LangGraph, AWS Bedrock
- **Web Development:** React, Node.js (MERN Stack), REST APIs, HTML/CSS
- **Cloud & DevOps:** Amazon Web Services (AWS) (EC2), Azure, Docker, Git, Terraform
- **Databases & Tools:** MongoDB, SQL, Databricks